Adaptive Fraud Detection with Dynamic Thresholding, Cost-Sensitive Learning, and OOD Detection

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***Abstract*—We present a complete fraud detection pipeline on the Kaggle Credit Card Fraud dataset (284,807 transactions; 0.17% fraud prevalence). Four supervised classifiers are trained and compared: Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), and Histogram-based Gradient Boosted Trees (GBT). Three class-imbalance strategies are evaluated: balanced class weighting, SMOTE, and random under-sampling. We introduce dynamic threshold tuning and a cost-sensitive framework (€500 FN / €10 FP). As a novel extension, we deploy Isolation Forest as an unsupervised out-of-distribution (OOD) detector trained solely on legitimate transactions, exploring its ability to flag anomalous — potentially novel — fraud patterns. RF with balanced class weights at threshold 0.192 achieves the best supervised performance: PR-AUC = 0.8116, Recall = 82.4%, Precision = 84.7% on the held-out test set. Code and notebook: https://github.com/omsureshprajapati/EE559-Fraud-Detection.**

***Index Terms*—fraud detection, class imbalance, SMOTE, threshold tuning, cost-sensitive learning, random forest, decision tree, out-of-distribution detection, isolation forest**

I. Introduction

Credit card fraud causes billions of euros in annual losses [1]. Automated detection is a binary classification problem with two fundamental challenges. First, the extreme class imbalance (577:1) makes accuracy useless as a metric. Second, error costs are asymmetric: a missed fraud (false negative, FN) costs far more than a false alarm (false positive, FP).

This paper addresses both challenges and introduces a third: *novel fraud*. Supervised models learn from labelled examples of known fraud patterns; they may fail to detect entirely new tactics that look unlike any training example. We address this with Isolation Forest [3], an unsupervised OOD detector trained only on legitimate transactions, which flags anomalous inputs regardless of whether they resemble known fraud.

We use PR-AUC and ROC-AUC as primary metrics [2], compare four supervised classifiers with three imbalance strategies, and introduce a cost-sensitive threshold selection procedure on the validation set.

II. Dataset and Preprocessing

A. Dataset

The Kaggle Credit Card Fraud Detection dataset [4] contains 284,807 European cardholder transactions from September 2013, with 492 fraud cases (0.173%). Features `V1`–`V28` are anonymised PCA components (zero-mean, decorrelated); `Time` and `Amount` (EUR) are in original units. No missing values; 1,081 duplicates retained.

B. Preprocessing Pipeline

The following strict ordering prevents data leakage:

1) **Split first.** Stratified 70/15/15 train/validation/test split (199,356 / 42,729 / 42,722), preserving 0.173% fraud in each.
2) **Scale on train only.** `StandardScaler` fitted on `Time` and `Amount` from training data; applied to all splits. `V1`–`V28` require no re-scaling.
3) **Resample train only.** SMOTE and under-sampling applied only within the training split.
4) **OOD detector on legitimate train only.** Isolation Forest fitted exclusively on the 199,012 legitimate training samples — no fraud labels used.

III. Methodology

A. Supervised Models

Logistic Regression (LR). Linear baseline with L2 regularisation ($C = 1.0$) and `class_weight='balanced'`, giving fraud samples ≈577× more influence in the loss.

Decision Tree (DT). Single decision tree with `class_weight='balanced'` and `max_depth=10` to limit overfitting. Fully interpretable: each prediction follows one path from root to leaf. Serves as a direct, interpretable reference against which the ensemble methods (RF, GBT) are compared.

Random Forest (RF). Ensemble of 100 trees, each trained on a bootstrap sample with random feature subsets [5]. Reduces variance via averaging. Same balanced class weighting.

Gradient Boosted Trees (GBT). `HistGradientBoostingClassifier` [7] (200 iterations), additively correcting residuals. Two variants: balanced class weighting (GBT-bal) and SMOTE-resampled training (GBT-SMOTE).

TABLE I: Resampling strategy comparison on LR (validation, threshold = 0.50).

Strategy	PR-AUC	Recall	F1
Class weight	0.7520	0.9189	0.1265
SMOTE	0.7599	0.9324	0.1209
Under-sampling	0.6986	0.9324	0.0865

B. Out-of-Distribution Detection

Isolation Forest [3] detects anomalies by recursively partitioning the feature space: anomalous samples require fewer splits to isolate because they occupy sparse regions. We train it on 199,012 legitimate training transactions only, with `contamination` set to the true fraud rate (0.0017). The anomaly score $s(x) = -\text{scoreSamples}(x)$ is higher for more anomalous transactions; a transaction is flagged as OOD if $s(x)$ exceeds the contamination-calibrated boundary.

This approach captures fraud patterns that are *anomalous relative to normal behaviour* — complementing supervised models that detect fraud patterns resembling the training label distribution.

C. Threshold Tuning and Cost-Sensitive Evaluation

We sweep $t \in [0.01, 0.99]$ (200 steps) on the validation set and find:

$$t^*_{\text{F1}} = \arg\max_t \text{F1}(t), \tag{1}$$

$$t^*_{\text{cost}} = \arg\min_t \left[\text{FN}(t) \cdot C_{\text{FN}} + \text{FP}(t) \cdot C_{\text{FP}}\right], \tag{2}$$

with $C_{\text{FN}} = 500\,€$ and $C_{\text{FP}} = 10\,€$.

IV. Results

A. Resampling Strategy Comparison

Table I compares three imbalance strategies on LR at threshold 0.50. Class weighting and SMOTE achieve similar PR-AUC; random under-sampling underperforms due to discarding 99.8% of legitimate training samples. SMOTE is applied to GBT as a secondary variant.

B. Model Comparison at F1-Optimal Threshold

Table II compares all supervised models at their F1-optimal thresholds on the validation set. RF achieves the best PR-AUC (0.864) and a strong F1 (0.877). DT achieves the same recall as LR (0.865) but with PR-AUC only 0.656 — significantly below both ensemble methods — confirming that averaging over 100 trees substantially reduces variance and improves calibration. GBT-SMOTE achieves the highest F1 (0.879) but at a lower PR-AUC (0.853) than RF, indicating it is more threshold-sensitive.

C. Threshold Tuning and Cost Analysis

Fig. 1 shows F1, Precision, and Recall vs. threshold. The DT curve confirms high instability relative to RF. Fig. 2 shows business cost vs. threshold. Because $C_{\text{FN}}/C_{\text{FP}} = 50$, cost-optimal thresholds are substantially lower than F1-optimal thresholds for all models (Table III). For RF, the cost-optimal threshold drops from 0.192 to 0.044, trading 37 additional FP for one fewer FN (€130 saving).

TABLE II: Supervised models at F1-optimal thresholds (validation). Best in **bold**.

Model	PR-AUC	ROC-AUC	Thr.	F1	Rec.	Prec.
LR (bal.)	0.752	0.986	0.990	0.727	0.865	0.628
DT (bal.)	0.656	0.945	0.990	0.727	0.865	0.628
RF (bal.)	**0.864**	0.951	**0.192**	0.877	**0.865**	0.889
GBT (bal.)	0.784	**0.982**	0.951	0.853	0.824	0.884
GBT (SMOTE)	0.853	0.973	0.960	**0.879**	0.838	**0.925**

TABLE III: F1-optimal vs. cost-optimal thresholds (validation set).

Model	F1-opt.	Cost-opt.
LR (balanced)	0.990	0.985
DT (balanced)	0.990	0.951
RF (balanced)	0.192	0.044
GBT (balanced)	0.951	0.837
GBT (SMOTE)	0.960	0.251

D. Final Test Set Evaluation

RF (balanced) was selected as the best model by validation PR-AUC = 0.864. Table IV shows test results; Fig. 3 shows the full evaluation. At the F1-optimal threshold, RF catches 61 of 74 frauds (82.4% recall) with only 11 false alarms (total cost: €6,610).

E. Out-of-Distribution Detection

Isolation Forest was trained on 199,012 legitimate training transactions with contamination = 0.0017. Table V compares it with RF on the test set. Fig. 4 shows the anomaly score distribution and PR curve.

Several important findings emerge. First, Isolation Forest achieves a high ROC-AUC (0.939) — comparable to RF — but a very low PR-AUC (0.091). This illustrates a key lesson about imbalanced data: ROC-AUC can be misleadingly high because it measures rank ordering, which can look good even with poor precision at the operating threshold. PR-AUC is a more honest metric when positives are rare (0.17%).

Second, Isolation Forest catches only 17 of 74 test frauds (23%), with 82 false alarms and a business cost of €29,320 — 4.4× higher than RF. As a standalone detector, it is not competitive.

Third, and most notably, **none of the 13 frauds missed by RF are caught by Isolation Forest**. This is a significant finding: the missed frauds are not anomalous relative to the legitimate distribution, confirming that they genuinely resemble normal transactions. They are borderline cases irreducible by the current feature set, not novel attack patterns that an OOD detector would have an advantage on.

V. Discussion

Decision Tree vs. Random Forest. DT achieves the same recall as LR (86.5%) at threshold 0.990 but lower PR-AUC (0.656 vs. 0.864 for RF). The 0.208 PR-AUC gap directly

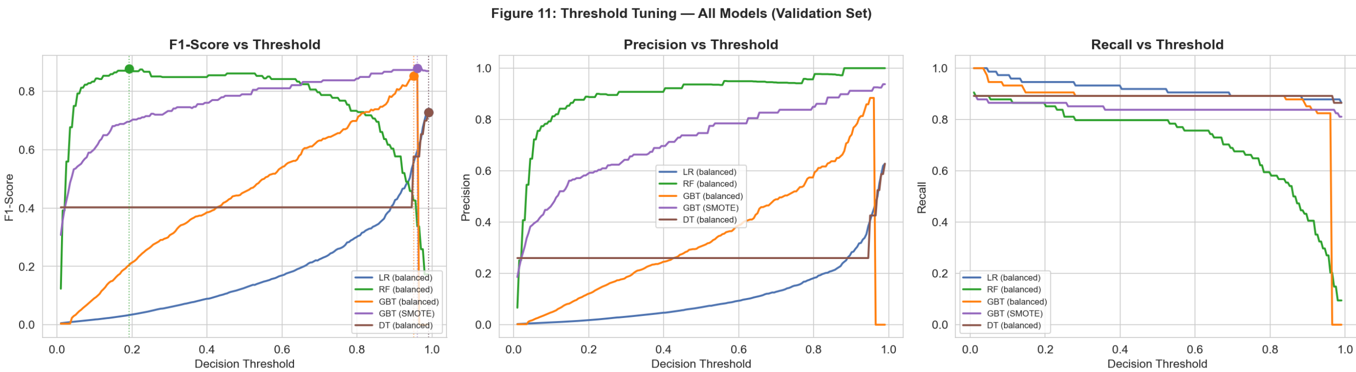

Fig. 1: F1-score, Precision, and Recall vs. threshold for all five supervised models (validation set). Dots mark the F1-maximising threshold. The DT curve shows higher instability than RF, illustrating the benefit of ensemble averaging.

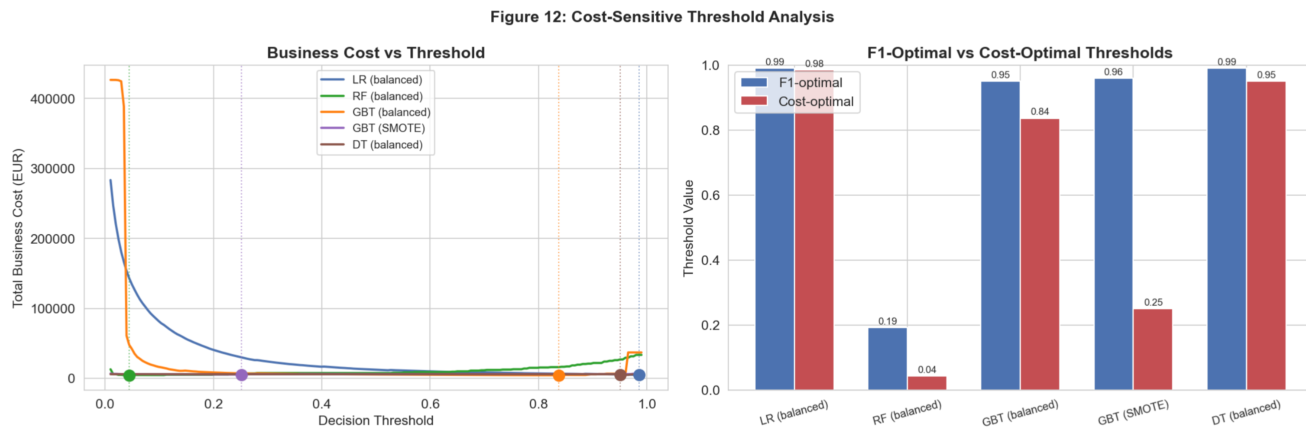

Fig. 2: Left: total business cost vs. threshold (validation set). Right: F1-optimal vs. cost-optimal thresholds — cost-optimal thresholds are consistently lower due to the 50:1 FN/FP cost ratio.

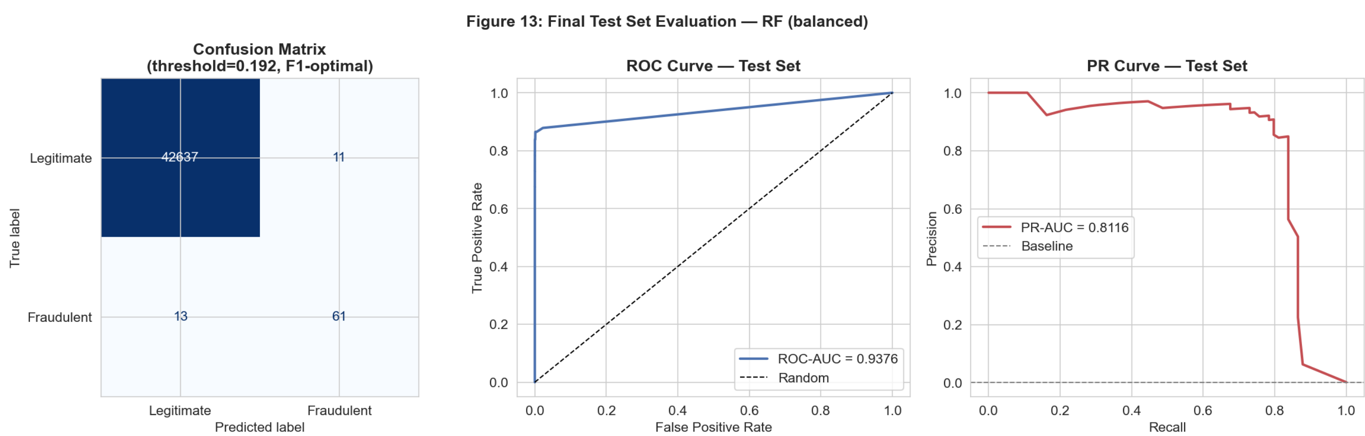

Fig. 3: RF (balanced) at threshold 0.192 on the held-out test set. Left: confusion matrix (61 TP, 11 FP, 13 FN, 42,637 TN). Centre: ROC curve (AUC = 0.9376). Right: PR curve (AUC = 0.8116).

quantifies the benefit of ensemble averaging: averaging 100 trees smooths out the noisy decision boundaries inherent to a single tree on a 577:1 imbalanced dataset. The DT result motivates using RF rather than a simpler model.

Supervised outperforms OOD when labels exist. Isolation Forest's PR-AUC (0.091) is 9× lower than RF's (0.812). This is expected: the model has no knowledge of what fraud looks like, only what legitimate looks like. However, the high ROC-AUC (0.939) confirms that fraud transactions are indeed anomalous in the feature space — just not anomalous *enough* to rank above

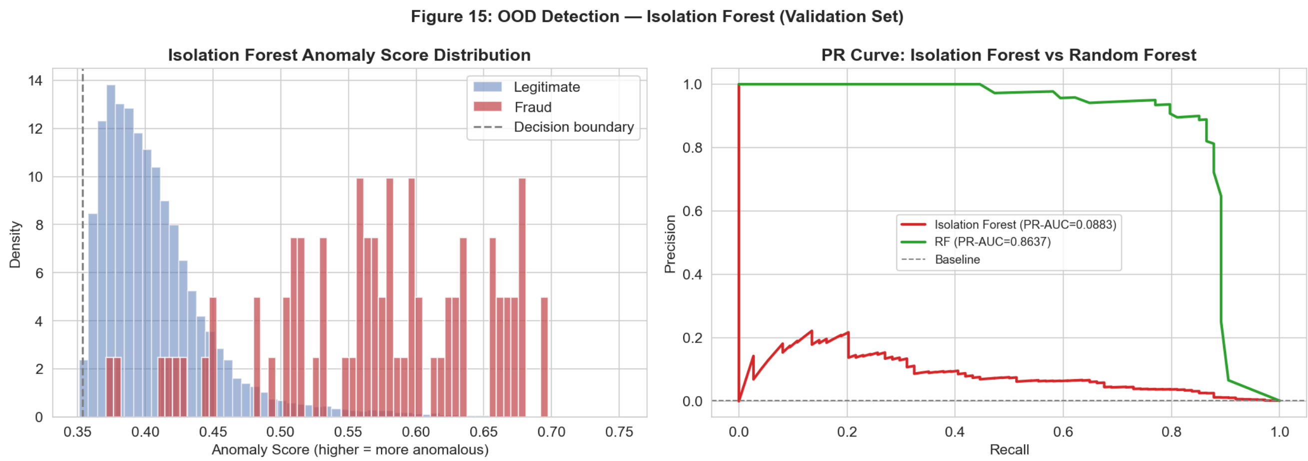

Fig. 4: Left: Isolation Forest anomaly score distribution for legitimate vs. fraud transactions (validation set). Right: PR curves for Isolation Forest vs. RF — the large gap confirms that supervised learning far outperforms unsupervised OOD detection when labelled fraud data is available.

TABLE IV: RF (balanced) on held-out test set (74 fraud; 42,648 legitimate).

Threshold	ROC	PR	F1	Rec.	Prec.	Cost (€)
0.192 (F1-opt.)	0.9376	0.8116	0.8356	0.8243	0.8472	6,610
0.044 (cost-opt.)	0.9376	0.8116	0.6739	0.8378	0.5636	6,480

TABLE V: Isolation Forest vs. RF (balanced) on the held-out test set.

Model	ROC	PR	Rec.	Prec.	Cost (€)
RF (t = 0.192)	0.9376	0.8116	0.8243	0.8472	6,610
Isolation Forest	0.9386	0.0905	0.2297	0.1717	29,320

legitimate ones at the 0.17% precision threshold required for high-precision detection.

OOD detection is most valuable when labels are absent or sparse. The practical implication is that Isolation Forest would be most useful in a *cold-start* scenario (no fraud labels yet) or for flagging genuinely novel attack patterns not seen in training. In a mature system with labelled fraud data, the supervised pipeline dominates.

Threshold tuning matters as much as model choice. Tuning RF's threshold from 0.50 to 0.192 improves validation F1 by 0.016 points with no model change. The cost-sensitive threshold (0.044) further reduces business cost by €130 vs. the F1-optimal operating point.

A. Limitations

Features `V1`–`V28` are opaque PCA components. Hyperparameter tuning was not performed. The 48-hour window limits temporal generalisation.

B. Future Work

Combining RF and Isolation Forest via a meta-classifier could improve recall on novel fraud without degrading precision. SHAP-based explanations [8], time-aware cross-validation, and online learning [9] for distribution shift are natural extensions.

VI. Conclusion

We presented a fraud detection pipeline comparing four supervised classifiers (LR, DT, RF, GBT), three imbalance strategies, dynamic threshold tuning, cost-sensitive evaluation, and a novel OOD extension via Isolation Forest. RF with balanced class weights at threshold 0.192 achieves the best result: PR-AUC = 0.8116, F1 = 0.8356, and business cost €6,610 on the test set. Decision Tree analysis confirms that ensemble averaging is essential for this task. The OOD experiment reveals a key insight: the frauds missed by RF are not detectable as anomalies either, indicating they are genuinely borderline cases rather than novel attack strategies — an important finding for practitioners deploying fraud detection systems.

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